

SCOTECI: Synthetic Chain-of-Thought Demonstrations for Document-Level Event Causality Identification

[TODO: working title — alternatives: “What Matters in Document-Scale Few-Shot ECI?” / “Rationalize, Select, Resample: Demonstration Design for Document-Level ECI”]

Abstract

- LLMs applied to Event Causality Identification (ECI) suffer from causal hallucination: near-total recall, unusable precision [Gao et al., 2023].
- We propose SCOTECI: LLM-based ECI at the *document* level — the full document text sits in the context and one call labels every pair jointly under a structured reasoning protocol with explicit coherence rules; demonstrations are whole labelled documents equipped with *synthetic chain-of-thought*: gold-guided reasoning generated by an LLM, then decontaminated so it reads as genuine discovery.
- Beyond the method, a systematic study of the demonstration design space: (i) CoT synthesis protocol (2-step gold-guided vs. 3-step blind-attempt-then-correct); (ii) demonstration selection (random, constant, TF-IDF, sentence-embedding, mention-set overlap); (iii) re-sampling with majority vote — number of passes and whether each pass sees the *same* or a *distinct* demonstration set; (iv) backbone choice across four contemporary LLMs.
- No fine-tuning, no human rationales, no external knowledge base.
- Headline: demonstration design dominates backbone choice — selection strategy and CoT protocol move F1 by [TODO: X] points where swapping backbones moves it by [TODO: Y]; on Causal-TimeBank, F1 [TODO: re-run on final backbone ladder] with balanced precision/recall instead of the degenerate high-recall regime.

1 Introduction

- **Task and stakes.** ECI = given a document and event mentions, decide for event pairs whether a (directed) causal relation holds [Cheng et al., 2025]. Feeds event forecasting, QA, narrative understanding.
- **The failure mode.** Prompted LLMs need no training data but exhibit *causal hallucination* [Gao et al., 2023]: recall of 80–96% with precision below 10% on sparse corpora — the model asserts causality wherever two events co-occur.
- **Idea 1 — operate at the document scale.** The full document text is placed in the context window and a single call labels *all* candidate pairs jointly, under an in-prompt protocol (mention pre-filter → per-pair for/against arguments → plausibility ranking → final JSON) plus explicit coherence rules (symmetry, dataset-specific transitivity). The model sees the whole narrative and produces the document’s causal graph in one coherent generation — deliberately betting on long-context reasoning quality, so the method *improves as models improve*.

- **Idea 2 — synthetic chain-of-thought few-shots.** Document-level demonstrations are whole labelled documents from the training split. Bare input→gold-JSON demos teach the output format, not the reasoning; human-written document-level rationales do not exist. We therefore *generate* them:
 - gold-guided generation: the demo document is presented under the exact task prompt together with its gold labels; the LLM writes reasoning that leads naturally to those labels;
 - decontamination: the reasoning is rewritten from scratch to purge any foreknowledge phrasing, so every sentence reads as discovery, not confirmation;
 - 3-step variant: blind attempt → gold-based correction → decontamination (preserves the texture of a genuine attempt);
 - injected as multi-turn (user, assistant) exchanges; cached per demo document.
- **Idea 3 — the demonstration design space is the lever.** With the formulation fixed, three orthogonal knobs remain: *how* the synthetic reasoning is produced (2- vs. 3-step), *which* documents are shown (five selection strategies from query-independent to semantically retrieved), and *how many* votes are taken with *what* demonstration diversity (resampling with shared vs. distinct demonstration sets per pass). We map this space systematically across four backbones.
- **Results preview.** Single evaluation table (Table 1): [TODO: headline numbers on the final backbone ladder — pilot runs give CTB F1 51.0 with P/R 53.2/49.0, i.e. the balanced regime, vs. single-digit precision for zero-shot prompting]. Ablations: [TODO: selection-strategy and CoT-depth deltas]; backbone choice is *not* the dominant factor [TODO: quantify].
- **Contributions.**
 - A document-level formulation of LLM-based ECI — full text in context, one generation labels every pair, structured protocol and in-prompt coherence rules, $O(1)$ calls per document.
 - A synthetic-CoT demonstration pipeline (gold-guided generation + decontamination, with a 3-step blind-then-correct variant), model-agnostic, requiring no human rationales and no knowledge base.
 - A systematic study of demonstration selection, CoT synthesis depth, and demonstration-varied resampling, showing that demonstration design outweighs backbone choice [TODO: confirm once the grid lands].

2 Related Work

2.1 LLMs and Causal Reasoning over Text

- Gao et al. [2023]: systematic evaluation; ChatGPT-class models over-predict causality (causal hallucination); established the preprocessing and evaluation protocol we follow.
- Surveys position ECI as a core event-semantics task with growing LLM interest [Cheng et al., 2025].

- Gap: prompting work reports either zero-shot or label-only few-shot results; the demonstration design space — what the demonstrations contain, how they are chosen, how they interact with resampling — is unmapped at document scale. That map is this paper.

2.2 In-Context Learning and Demonstration Selection

- ICL emerges with scale [Brown et al., 2020] and is highly sensitive to demonstration choice and content [Liu et al., 2022, Min et al., 2022].
- Retrieval-based selection (lexical or embedding similarity to the query) is the standard recipe [Liu et al., 2022]; whether it holds when each demonstration is an entire labelled document — long, expensive, and few — is one of our questions (§6.2).

2.3 Chain-of-Thought, Synthetic Rationales, and Resampling

- CoT prompting boosts multi-step reasoning [Wei et al., 2022]; Auto-CoT generates rationales with zero-shot-CoT over clustered queries [Zhang et al., 2023]; STaR bootstraps rationales via *rationalization* — generating reasoning conditioned on the known answer — but consumes them as fine-tuning data [Zelikman et al., 2022]; step-by-step distillation extracts rationales as training signal [Hsieh et al., 2023].
- Self-consistency samples multiple reasoning paths and majority-votes the answers [Wang et al., 2023]; the paths vary by sampling temperature. Our resampling instead varies the *evidence*: at temperature 0, each pass can receive a different pair ordering and, optionally, a different demonstration set — prompt-side rather than decoder-side diversity.
- Ours differs: rationalization used *at inference time* as ICL demonstrations, at document scale, with an explicit decontamination pass to remove answer-leakage phrasing — without which the demonstrations teach the model to assert rather than reason.

3 Task Formulation

- Input: document D with inline-marked event mentions $M = \{m_1, \dots, m_n\}$ (<ID word> format), and a pair set P : the annotated candidate list when the corpus provides one (constrained setting), otherwise all unordered mention pairs (open-ended setting).
- Output: a complete labelling $\ell : P \rightarrow \{\textit{causes}, \textit{causedby}, \textit{norel}\}$ — a unified directed label space into which each corpus’ native labels are mapped [TODO: spell out per-dataset label mapping (CAUSE/PRECONDITION, FALLING_ACTION, CauseEffect/EffectCause)].
- Document-scale inference: the *entire* document text is in the context — no windowing, no chunking — and the model produces ℓ for *all* of P in a single generation, as a JSON array. Every labelling decision is made with the full narrative visible, so cross-sentence chains, confounders, and discourse structure are available to every pair judgement simultaneously.
- Evaluation: micro precision/recall/F1 over the positive (causal) classes; *norel* excluded; intra- vs. inter-sentence subsets where the corpus distinguishes them; preprocessing follows Gao et al. [2023].

4 Method

4.1 Overview

- Pipeline (Figure ??): test document $\rightarrow$ select k demonstration documents from the training split $\rightarrow$ generate (and cache) synthetic CoT for each $\rightarrow$ assemble multi-turn prompt (demonstrations as past exchanges, then the test document) $\rightarrow$ document-level LLM call(s) $\rightarrow$ parse JSON $\rightarrow$ optional resampling with majority vote.
- Four components, each carrying a studied design dimension: the document-level structured prompt (§4.2), demonstration selection (§4.3), synthetic-CoT generation with its 2- and 3-step protocols (§4.4), and resampling with shared or distinct demonstration sets (§4.5).

4.2 Document-Level Structured Prompting

- One call per document; the full text in context; every pair of P requested at once; the model sees the whole narrative, not a windowed snippet.
- In-prompt reasoning protocol (four ordered steps, enforced by the system prompt):
 - Step 1 — mention-level pre-filter: affirmatively argued exclusions (“ m is a location with no agentive role”), never blanket dismissal; surviving mentions listed.
 - Step 2 — per-pair arguments *for* (connective, mechanism, counterfactual asymmetry, discourse structure) and *against* (temporal-only, confounder, reporting verb, distance).
 - Step 3 — qualitative ranking of retained pairs by causal plausibility, no labels committed yet.
 - Step 4 — final JSON over *every* requested pair; everything pre-filtered defaults to *norel*.
- Precision-first rule block (targets causal hallucination directly): two-signal rule for non-overt links; stricter gate for inter-sentential / long-range pairs; default-to-*norel* under uncertainty; confounder / proximal-cause check; reporting and temporal connectives explicitly ruled non-causal; counterfactual direction test.
- In-prompt coherence rules: symmetry ($causes(i, j) \Leftrightarrow causedby(j, i)$) and dataset-specific transitivity rules — checkable at parse time because the whole graph is produced jointly [TODO: decide whether the coherence checker/post-hoc rule pruning appears in main text or appendix].
- Why this leverages better models: the protocol asks for exactly the global, long-context reasoning that frontier models are differentially good at.

4.3 Demonstration Selection

- Demonstration pool: training split of the same corpus (capped pool size to bound CoT-generation cost) [TODO: final pool size].
- Demonstrations are *whole documents with their complete gold labelling*, including the *norel* pairs — every demo carries explicit negative evidence alongside the positives.
- Five selection strategies, spanning query-independent to semantically adaptive (all evaluated in §6.2):

- *random* — k documents drawn uniformly per test document; query-independent and varying;
 - *constant* — one fixed set of k documents reused for every test document (chosen once, e.g. the most compact well-annotated documents in the pool); query-independent and frozen — isolates whether *adaptivity* matters at all, and maximizes prompt-cache reuse;
 - *TF-IDF* — cosine over full document text, top- k ; lexical similarity;
 - *sentence-embedding* — cosine over pooled sentence-transformer embeddings of the document text, top- k ; semantic similarity [TODO: state embedding model];
 - *mention-set overlap* — Jaccard similarity between the *event mention sets* of test and candidate documents: two documents that share many event types (“attack”, “collapse”, “rescue”) likely exercise similar causal schemas, independently of surface wording.
- Default configuration: TF-IDF, $k = 3$ [TODO: confirm after the selection ablation].

4.4 Synthetic Chain-of-Thought Generation

- Motivation: label-only demos teach the output schema, not the reasoning; document-level human rationales are unavailable and prohibitively expensive to write; gold labels for training documents are available — so rationales can be *synthesized*.
- 2-step protocol, per demonstration document:
 - (i) *gold-guided generation*: the demo document is shown under the exact task prompt, plus its gold labels and the instruction to write reasoning that “leads naturally to these labels from the text alone” without revealing foreknowledge;
 - (ii) *decontamination*: the response is rewritten from scratch to purge leakage phrasing (“the labels show. . .”, “as indicated by the expected output. . .”), so each sentence reads as analysis, not confirmation.
- 3-step protocol: (i) *blind attempt* with no gold access — the model reasons over the demo document exactly as it would at test time; (ii) *gold-based correction* that fixes wrong conclusions while keeping correct reasoning; (iii) decontamination. Hypothesis: the blind attempt preserves the texture, hedging, and difficulty profile of genuine inference — reasoning the test-time model can actually imitate — where gold-guided generation may read as too fluent, arguing every pair with unearned confidence. Costs one extra generation call per demo. The 2- vs. 3-step comparison is a primary ablation (§6.1).
- Injection: each demo becomes a (user task turn, assistant CoT turn) exchange preceding the test document — the model “remembers having done” k documents correctly, reasoning included; compatible with multi-step task prompts (one synthetic response per step).
- Generated once per (document, protocol), cached across the run and on disk across runs; generator model = inference model by default (self-generated rationales) [TODO: teacher-vs-self generation experiment — does a stronger generator lift a weaker reasoner?].
- Relation to STaR-style rationalization [Zelikman et al., 2022]: same trick — condition on the answer to obtain reasoning — but used as in-context demonstrations rather than fine-tuning data, and with an explicit decontamination pass, which we show is load-bearing (§6.1).

4.5 Resampling and Aggregation

- Temperature 0; strict JSON parsing with bounded retries; missing pairs default to *norel*.
- Resampling: N independent passes with per-pair majority vote, ties broken to *norel* (precision-preserving). Because decoding is greedy, diversity is injected on the *prompt side*:
 - *pair-order reshuffle*: the candidate pair list is re-ordered per pass (deterministically seeded per document and pass index, so runs are reproducible) — breaks position bias;
 - *shared demonstrations* (baseline): every pass sees the same k demos — votes differ only through pair order;
 - *distinct demonstrations*: $N \times k$ documents are selected by the active strategy and split round-robin across passes (each pass gets a rank-balanced set of k) — every vote is grounded in different evidence, decorrelating errors that stem from an unlucky demonstration draw. Requires a pool of at least $N \times k$ eligible documents.
- The resampling study (§6.3) crosses $N \in \{1, 3, 5\}$ with $\{\text{shared}, \text{distinct}\}$ demonstrations: does voting help, and does demonstration diversity add anything beyond pair-order diversity? [TODO: run the grid].
- Cost: N calls per document plus $2\text{--}3k$ amortized CoT-generation calls (cached); distinct demonstrations multiply CoT generation by N but not test-time calls; quantified in §6.5.

5 Experiments

5.1 Datasets and Metrics

- EventStoryLine (ESC) [Caselli and Vossen, 2017]: 22 topics, 258 docs, 1,770 causal pairs [TODO: confirm version used (0.9 vs 1.5) and align statement with Gao preprocessing].
- Causal-TimeBank (CTB) [Mirza et al., 2014]: 183 docs, 318 causal pairs — sparsest positives, hence the dataset where hallucination is most punishing; no intra/inter split.
- MAVEN-ERE [Wang et al., 2022]: large-scale Wikipedia corpus, 57,992 causal pairs [TODO: state evaluated subset/sample size].
- Preprocessing follows Gao et al. [2023]; micro P/R/F1 on the causal classes, *norel* excluded.
- [TODO: decide whether the MECI multilingual evaluation [Lai et al., 2022] goes in (appendix?) — pipeline supports it.]

5.2 Backbones and Baselines

- Four contemporary backbones spanning tiers and architectures, all via a unified API [TODO: exact endpoint identifiers]:
 - GPT-5.5 (frontier closed);
 - GLM-5.2 (large open);
 - DeepSeek-V4 (large open MoE);
 - Qwen3-30B-Thinking (mid-size open, explicit reasoning mode).

- Baselines, same backbone and same document-scale setting so every comparison is within-formulation:
 - *Base*: task description + document + pair list, direct JSON answer;
 - *CoT* [Wei et al., 2022]: step-by-step reasoning requested before the JSON;
 - SCOTECI: full pipeline (structured protocol + synthetic-CoT demonstrations + resampling) [TODO: state final config used in the main table].
- Model comparison is a *secondary* axis by design: the paper’s claim is that demonstration design, not backbone choice, is the dominant lever [TODO: verify the claim survives the grid].

5.3 Implementation Details

- $k = 3$ demonstrations, TF-IDF selection, 2-step CoT protocol, temperature 0, resampling $N = 3$ with distinct demonstration sets [TODO: confirm final default config; pool size; exact prompt files versioned in the repo].
- Synthetic CoT generation cost: 2 (or 3) extra calls per unique demo document, amortized by caching [TODO: give measured token counts].
- Deterministic pipeline: fixed evaluation subset (first n documents of each split), per-document seeded pair-order shuffling, per-(document, pass) seeded resample reshuffling — identical runs are exactly reproducible up to provider nondeterminism.

5.4 Main Results

- **CTB (expected headline).** Pilot runs put SCOTECI in the balanced regime ($P \approx R \approx 50$) where zero-shot prompting is degenerate (single-digit precision at 80–96 recall, Gao et al., 2023) [TODO: final numbers per backbone].
- **Reading across method rows.** Base/CoT expected in the high-recall/low-precision regime; SCOTECI trades recall for large precision gains — quantify the P/R rebalancing per dataset [TODO: once the grid lands].
- **Reading across backbone blocks.** The between-backbone spread within a method row vs. the between-method spread within a backbone block — the central comparison for the “demonstration design dominates backbone choice” claim [TODO: compute both spreads].
- [TODO: add significance / resampling variance statement once final runs land.]

6 Analysis

6.1 Synthetic CoT: Protocol Depth and Decontamination

- Remove synthetic CoT entirely (demos = document + gold JSON only): isolates the contribution of *reasoning-bearing* demonstrations [TODO: expected: format retained, precision drops — run].

Table 1: Main results: micro P/R/F1 on the causal classes. All rows use the same document-scale setting (full text in context, all pairs labelled in one generation); SCOTECI adds synthetic-CoT demonstrations and resampling. [TODO: fill after the final grid; mark best per column in bold; state resampling config; note any intra-sentence-only caveats].

Method	ESC			CTB			MAVEN-ERE		
	P	R	F1	P	R	F1	P	R	F1
GPT-5.5:									
Base	–	–	–	–	–	–	–	–	–
CoT	–	–	–	–	–	–	–	–	–
SCOTECI	–	–	–	–	–	–	–	–	–
GLM-5.2:									
Base	–	–	–	–	–	–	–	–	–
CoT	–	–	–	–	–	–	–	–	–
SCOTECI	–	–	–	–	–	–	–	–	–
DeepSeek-V4:									
Base	–	–	–	–	–	–	–	–	–
CoT	–	–	–	–	–	–	–	–	–
SCOTECI	–	–	–	–	–	–	–	–	–
Qwen3-30B-Thinking:									
Base	–	–	–	–	–	–	–	–	–
CoT	–	–	–	–	–	–	–	–	–
SCOTECI	–	–	–	–	–	–	–	–	–

- 2-step (gold-guided → decontaminate) vs. 3-step (blind attempt → correct → decontaminate): does preserving the texture of a genuine attempt beat fluent gold-guided reasoning? Does the answer depend on the backbone (a stronger model may produce blind attempts worth correcting; a weaker one may not)? [TODO: run both protocols across the backbone ladder — this is the headline ablation].
- Decontamination on/off: does leakage phrasing in demos teach the model to assert without reasoning? [TODO: run — this justifies the step].
- Qualitative: side-by-side of a 2-step and a 3-step rationale for the same demo document [TODO: pick a CTB doc; appendix if long].

6.2 Demonstration Selection

- Grid: {random, constant, TF-IDF, sentence-embedding, mention-set overlap} × datasets, default config otherwise [TODO: run].
- Hypotheses to test:
 - adaptivity gap: do query-dependent strategies (TF-IDF, embedding, mention overlap) beat query-independent ones (random, constant) when each demo is an entire document?
 - lexical vs. semantic vs. event-level: mention-set overlap matches on *event vocabulary* rather than surface text — if causal schemas transfer along event types, it should win on corpora with recurring scenarios;

- constant as the cache-friendly floor: if constant is within noise of retrieval, the entire selection machinery can be dropped for a fixed cached prefix [TODO: cost implications in §6.5].
- Number of demonstrations $k \in \{1, 3, 5\}$ [TODO: run; doc-level demos are long, context dilution may bite sooner than in short-form ICL].

6.3 Resampling: Vote Count and Demonstration Diversity

- Grid: $N \in \{1, 3, 5\} \times \{\text{shared}, \text{distinct}\}$ demonstration sets, default config otherwise [TODO: run].
- Questions:
 - marginal value of votes: F1 vs. N — where does majority vote saturate at temperature 0, given that pair-order reshuffling is the only variance source in the shared setting?
 - does demonstration diversity add signal? Distinct sets decorrelate demonstration-induced errors: an unlucky demo hurts one vote, not all N . Expected to matter most for high-variance selection (random) and least for constant [TODO: test the interaction];
 - precision mechanics: ties break to *norel*, so voting is precision-preserving by construction — decompose the F1 movement into P and R components [TODO: compute].
- Practical note: distinct demonstrations need $N \times k$ eligible pool documents and multiply CoT-generation (not test-time) cost by up to N [TODO: measured overhead].

6.4 Backbone Comparison

- Deliberately last: rank the four backbones under the full method and under Base — if the method compresses the between-model gap, the demonstrations are doing work the raw model quality otherwise would [TODO: compute; figure F1-per-backbone under Base vs. SCOTECI, not a table].
- Reasoning-mode interaction: Qwen3-30B-Thinking emits its own long reasoning — do synthetic CoT demonstrations compound with or substitute for native thinking? [TODO: inspect traces; brief prose].
- Cross-model demo reuse: demos generated by one backbone consumed by another [TODO: optional; run only if time permits].

6.5 Cost Analysis

- LLM calls per document: N test-time calls plus 2–3 CoT-generation calls per unique demo document (cached across the run and across passes in the shared setting) [TODO: measured token counts; cost-vs-F1 figure across configs — constant selection + shared demos as the cheap end, embedding selection + distinct demos as the expensive end].

6.6 Error Analysis

- False-positive taxonomy: transitive flattening ($A \rightarrow B \rightarrow C$ read as $A \rightarrow C$ when the scheme does not credit it), temporal-as-causal, reporting verbs [TODO: sample and count from traces].

- Dense documents: does precision degrade with the number of gold pairs per document? [TODO: scatter/figure].
- Direction errors: *causes* vs. *causedby* confusions as a fraction of true-pair errors [TODO: compute from logs].
- Qualitative: one worked example — synthetic CoT demo, model reasoning, prediction vs. gold [TODO: pick a CTB doc].

7 Conclusion

- Formulated LLM-based ECI at the document scale: full text in context, a single structured generation labels every pair, coherence rules enforced in-prompt.
- Introduced synthetic, decontaminated chain-of-thought demonstrations: rationalization at inference time, no human rationales, no knowledge base, no fine-tuning — with a 2-step and a 3-step synthesis protocol.
- Mapped the demonstration design space — synthesis depth, five selection strategies, resampling count $\times$ demonstration diversity — across four backbones; demonstration design is the dominant lever, backbone choice secondary [TODO: confirm].
- Large precision-driven gains where hallucination is costliest [TODO: final CTB numbers]; balanced P/R elsewhere.
- Future work: multilingual MECI, per-document strategy routing, teacher-generated CoT for small models.

Limitations

- Some results are currently intra-sentence only (ESC, MAVEN-ERE); full-document numbers must land before the cross-dataset story is complete [TODO: track].
- Synthetic CoT costs 2–3 generation calls per demonstration document (amortized by caching, but nonzero); distinct-demonstration resampling multiplies the number of unique demos by up to N .
- Decontamination is itself LLM-based; residual leakage phrasing cannot be ruled out [TODO: manual audit of n sampled demos].
- Long documents with large pair sets stress the single-call format (output-length limits, attention dilution); per-document routing or chunking may be needed at the extreme.
- Provider-side nondeterminism means temperature-0 runs are reproducible in inputs but not bit-identical in outputs.

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